

GE Drive Systems

INSTRUCTIONS

FUSE AND SHUNT INTERCHANGE FOR HORSEPOWER MODIFICATION

QIII DC Controller "G", "C", AND "D" Frames

IMPORTANT
INFORMATION

CAUTION: To ensure proper operation of the QIII DC Controller, these instructions must be followed when changing fuses and/or shunts to modify the horsepower of the controller.

These instructions do not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation or maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to General Electric Company.

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SAFETY PRECAUTIONS

WARNING - SHOCK AND BURN HAZARD:
Always disconnect power to the controller before changing any components of the controller. Failure to do so may cause serious injury to personnel and damage to the controller or the controlled equipment.

CAUTION: Lexan insulation barrier must always be placed between 104X149CC032, 033, and 043 shunt modules and controller frame when mounting, then mounting screw cavities must be filled with RTV silicon rubber after mounting (for electrical insulation).

CAUTION: Always position brass washer (N401P41 or N401P43) on **top** of the voltage feedback wire connector on DC fuse as it is furnished to improve conductivity. If customer wiring is to be connected below the fuse, it must be positioned on **top** of the brass washer.

NOTE: *Read all WARNINGS, CAUTIONS, and NOTES in the QIII DC Controller Instruction Book (GEK-45167) prior to changing any components.*

NOTE: *Read and heed all WARNING, CAUTION, and NOTE labels posted on the equipment prior to changing any components.*

GENERAL DESCRIPTION

The QIII DC Controllers ("G", "C", and "D" Frame modules) are shipped from the factory with AC Fuses, DC Fuse (regenerative controllers only), and Current Shunt installed that are standard for a specific Module Catalog Number (see Table 1).

It may be necessary to change these components to tailor the controller to individual applications. The Programming Software Package will designate the proper components to be used from the controller application information entered when programming the controller. Use the listing of components provided by the Programming Software Package and these instructions to change components to the required combination.

NOTE: *When changing shunts and fuses, pay special attention information that pertains to quantity required and any additional hardware that may be required for the change.*

TABLE 1: QIII DC CONTROLLER STANDARD MODULE COMPONENTS "G", "C", AND "D" FRAMES

Module Catalog Number	Frame Size	AC Fuses (Qty. 3) 104X109	DC Fuse (Regen.) 104X109	Current Shunt 104X149	Blower Fuse* (If Used, Qty.3) 104X109
7VQD14	G	AG007	----	CC032	-----
7VQD14R	G	AG007	AJ007	CC032	-----
7VQD43	G	AG007	----	CC035	-----
7VQD43R	G	AG007	AJ007	CC035	-----
7VQD127	G	AG007	----	CC037	-----
7VQD127R	G	AG007	AJ009	CC037	-----
7VQD260	C	AG021	----	CA009	-----
7VQD260R	C	AG021	AJ015	CA009	-----
7VQD440	C	AG027	----	CA010	AD073
7VQD440R	C	AG027	AJ019	CA010	AD073
7VQD525	C	AG027	----	CA011	AD073
7VQD525R	C	AG027	AJ025	CA011	AD073
7VQD855	D	AG032	----	CC007	-----
7VQD855R	D	AG032	AJ026	CC007	-----
7VQD1285	D	AG032	----	CC002	AD038
7VQD1285R	D	AG032	AJ025 (Qty. 2)	CC002	AD038

Fuses listed are for 460 VAC applications.* *"C" Frame*** *230 VAC applications use fuse catalog #104X109AD038;* ***"D" Frame*** *230 VAC applications use fuse catalog #104X109AD080.*

"G" FRAME HORSEPOWER MODIFICATIONS

Shunt Changes - 104X149CC032 to CC033 or CC043 (See Figure 1)

NOTE: *These shunts are premounted modules that are secured directly to the controller frame with #8 thread-forming screws. When changing shunts, change the entire module.*

WARNING - SHOCK AND BURN HAZARD:

Always disconnect power to the controller before changing any components of the controller. Failure to do so may cause serious injury to personnel and damage to the controller or the controlled equipment.

CAUTION: Lexan insulation barrier must always be placed between 104X149CC032, 033, and 043 shunt modules and controller frame when mounting, then mounting screw cavities must be filled with RTV silicon rubber after mounting (for electrical insulation).

1. Verify that power to controller that shunt is to be changed is **OFF**.

2. Remove electrical connections from shunt as follows:
   a. Remove **control** connections by removing the two #10 screws with a flat-bladed screwdriver.
   b. Remove **power** connections by removing the two 1/4" x 20 bolts with a 7/16" wrench or socket.

3. Remove RTV from mounting screw cavity with a sharp instrument until screw head is accessible.

4. Remove the two mounting screws securing the shunt module to the controller frame and lift module and Lexan insulation barrier from frame.

5. Place Lexan barrier behind new shunt module (per Programming Software Package) to be mounted and position barrier and module over mounting holes in controller frame.

6. Place a #8 thread-forming screw in each mounting hole and tighten to 28 lb/in. to secure shunt module to controller frame.

7. Fill mounting screw cavity with RTV (for electrical insulation).

8. Replace electrical connections to shunt as follows:
   a. Orient all connectors to new shunt in same position as when removed in step 2 from old shunt.
   b. Place #10 brass screw (with lockwasher and flatwasher) through each control wire connector, thread screw into shunt and tighten firmly.
   c. Place 1/4" x 20 bolt with flatwasher through power cable connector, thread bolt into shunt, and tighten to 38-50 lb/in. with a 7/16" wrench or socket.
   d. Place 1/4" x 20 bolt with flatwasher into other tapped hole of shunt and tighten to 38-50 lb/in. with a 7/16" wrench or socket.

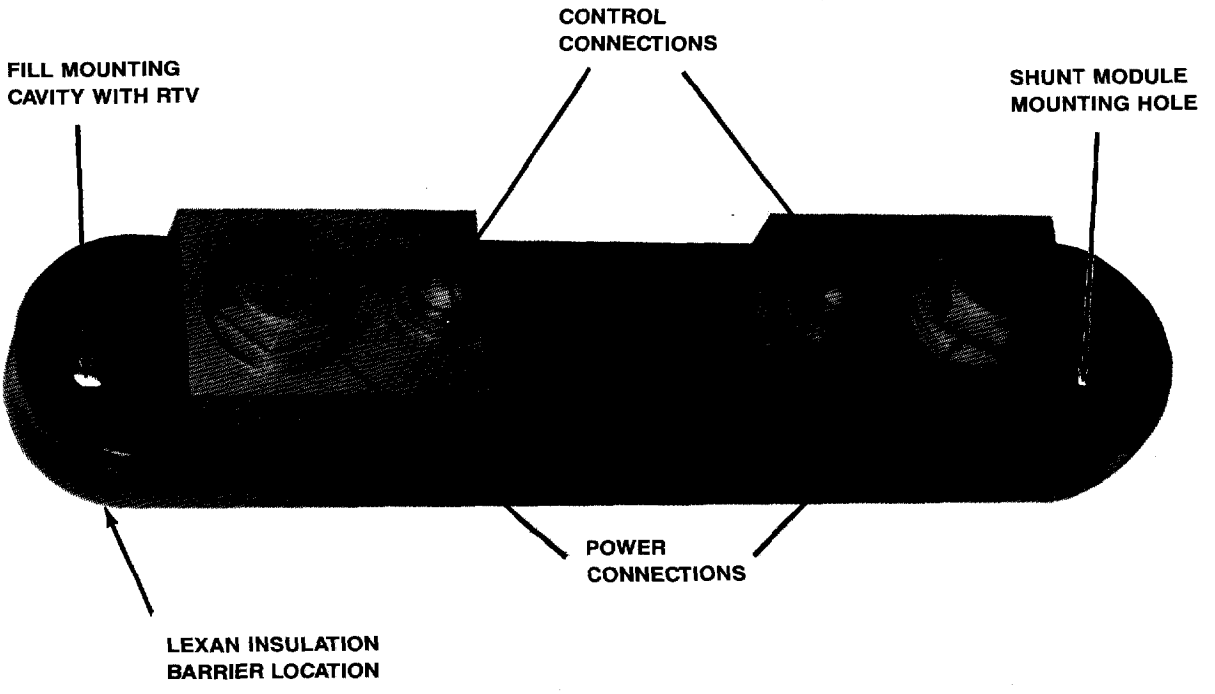


Figure 1. SHUNT MODULE ("G" FRAME)

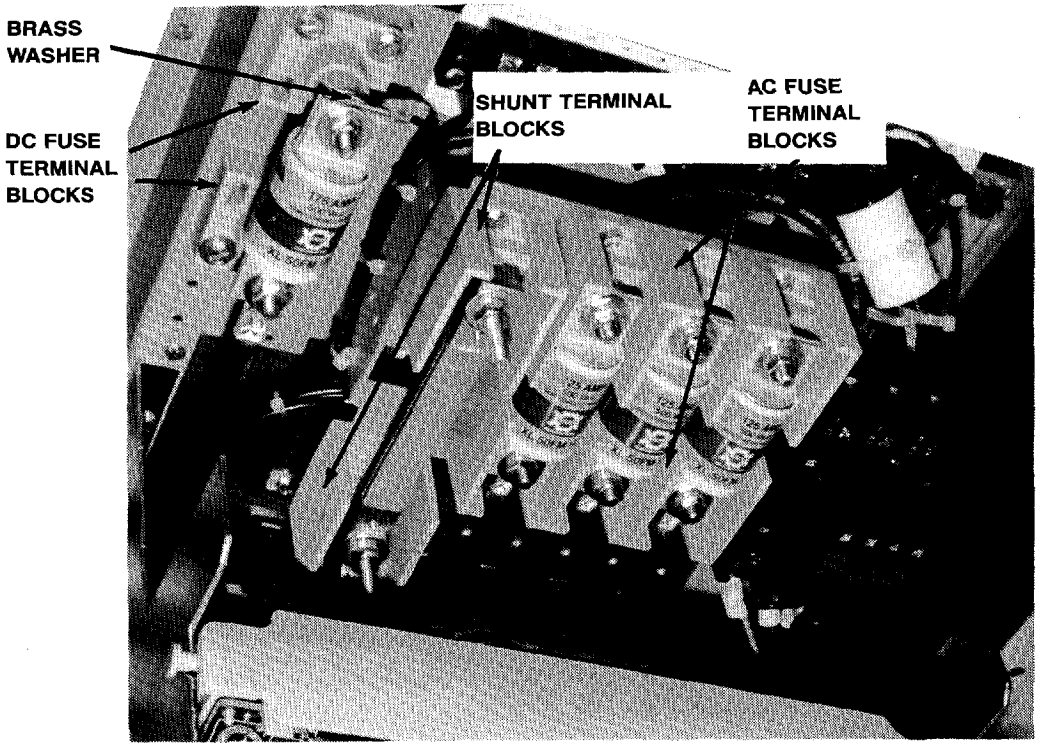


Figure 2. TYPICAL "G" FRAME SHUNT, FUSES, AND TERMINAL BLOCKS

Shunt Changes - 104X149CC032 *to* CC034 - CC040 and CC042 (See Figures 1 and 2)

NOTE: *When changing to these shunts, two shunt mounting terminal blocks (36B605553AAG15, includes hardware) must be mounted to the controller frame.*

WARNING - SHOCK AND BURN HAZARD:

Always disconnect power to the controller before changing any components of the controller. Failure to do so may cause serious injury to personnel and damage to the controller or the controlled equipment.

1. Verify that power to controller that shunt is to be changed is **OFF**.

2. Remove electrical connections from shunt as follows:
   a. Remove control connections by removing the two #10 screws with a flat-bladed screwdriver.
   b. Remove power connections by removing the two 1/4" x 20 bolts with a 7/16" wrench or socket.

3. Remove RTV from mounting screw cavity with a sharp instrument until screw head is accessible.

4. Remove the two mounting screws securing the shunt module to the controller frame and lift module and Lexan insulation barrier from frame.

5. Retrieve terminal blocks and mounting hardware and place a flatwasher over stud extending from each terminal block.

6. Orient and mount terminal blocks to controller frame as follows:
   a. Place new shunt (per Programming Software Package) over studs of terminal blocks and orient terminal blocks with mounting slots extending outward from ends of shunt.
   b. Position this assembly on frame and determine mounting hole in frame to be used (screws to be placed in outermost mounting holes only).
   c. Place a #8 thread-forming screw in outermost mounting holes of each terminal block and tighten to 28 lb/in. to secure terminal blocks to controller frame.

7. Replace electrical connections to shunt as follows:
   a. Orient all connectors to new shunt in same position as when removed in step 2 from old shunt.
   b. Place #10 brass screw (with lockwasher and flatwasher) through each control wire connector, thread screw into shunt, and tighten firmly.
   c. Place power cable connector over shunt mounting stud and then place a flatwasher, lockwasher, and 1/4" x 20 nut onto stud and tighten to 38-50 lb/in. with a 7/16" wrench or socket.
   d. Place a flatwasher, lockwasher, and 1/4" x 20 nut onto other mounting stud and tighten to 38-50 lb/in. with a 7/16" wrench or socket.

Shunt Changes - 104X149CC035 or 037 to CC034 - CC040 and CC042 (See Figure 2)

WARNING - SHOCK AND BURN HAZARD:

Always disconnect power to the controller before changing any components of the controller. Failure to do so may cause serious injury to personnel and damage to the controller or the controlled equipment.

1. Verify that power to controller that shunt is to be changed is **OFF**.

2. Remove electrical connections from shunt as follows:
   a. Remove **control** connections by removing the two #10 screws with a flat-bladed screwdriver.
   b. Remove **power** connections by removing the two 1/4" x 20 nuts, lockwashers, and flatwashers from the power cable mounting studs with a 7/16" wrench or socket.

3. Place new shunt (per Programming Software Package) over studs of terminal blocks (make sure flatwashers are still present under the shunt).

4. Replace electrical connections to shunt as follows:
   a. Orient all connectors to new shunt in same position as when removed in step 2 from old shunt.
   b. Place #10 brass screw (with lockwasher and flatwasher) through each control wire connector, thread screw into shunt, and tighten firmly.
   c. Place power cable connector over shunt mounting stud and then place a flatwasher, lockwasher, and 1/4" x 20 nut onto stud and tighten to 38-50 lb/in. with a 7/16" wrench or socket.
   d. Place a flatwasher, lockwasher, and 1/4" x 20 nut onto other mounting stud and tighten to 38-50 lb/in. with a 7/16" wrench or socket.

AC and DC Fuse Changes

NOTE: *When changing* ***DC fuse*** *to 104X109AJ014 or AJ015, the two fuse mounting terminal blocks must be changed to ones with 3/8" mounting studs (799C914ABG01, includes hardware) and a brass washer (N401P43) must also be used when connecting voltage feedback wire.*

WARNING - SHOCK AND BURN HAZARD:

Always disconnect power to the controller before changing any components of the controller. Failure to do so may cause serious injury to personnel and damage to the controller or the controlled equipment.

1. Verify that power to controller that fuse(s) is to be changed is **OFF**.

2. Remove electrical connections from fuse by removing the two 1/4" x 20 nuts with a 7/16" inch wrench or socket.

3. Remove lockwashers, flatwashers, and power cable from studs and lift off fuse to be replaced.

4. Position new fuse (per Programming Software Package) over mounting studs to determine if terminal blocks must be moved (or changed for 104X109AJ014 or AJ015 fuses) and proceed as follows: (if no movement or change is needed, proceed to step 5)

   If terminal blocks must be moved -

   a. Loosen and remove the #8 thread-forming mounting screws securing terminal blocks to frame and place new fuse onto the studs of terminal blocks (move blocks to accept new fuse).
   b. Position this assembly on frame to determine new mounting holes to be used (screws to be placed in outermost mounting holes only).
   c. Place a #8 thread-forming screw in outermost mounting holes of each terminal block and tighten to 28 lb/in. to secure terminal blocks to controller frame.

   If terminal blocks must be changed -

   a. Loosen and remove the #8 thread-forming mounting screws securing terminal blocks to frame and remove these terminal blocks.
   b. Retrieve new terminal blocks and mounting hardware and place a flatwasher over stud extending from each terminal block.

c. Place new fuse over studs of new terminal blocks and orient terminal blocks with mounting slots extending outward from ends of shunt.

d. Position this assembly on frame and determine mounting holes in frame to be used (screws to be placed in outermost mounting holes only).

e. Place a #10 thread-forming screw in outermost mounting holes of each terminal block and tighten to 55 lb/in. to secure terminal blocks to controller frame.

5. Make electrical connections to fuse as follows:

If terminal blocks were not changed -

a. Place power cable connector over fuse mounting stud from which it was removed in step 3, then place a flatwasher, lockwasher, and 1/4" x 20 nut onto stud and tighten to 38-50 lb/in. with a 7/16" wrench or socket.

b. Place flatwasher, lockwasher, and 1/4" x 20 nut onto other mounting stud and tighten to 38-50 lb/in. with a 7/16" wrench or socket (voltage feedback wire and brass washer should still be present if DC fuse).

If terminal blocks were changed -

a. Place power cable connector over fuse mounting stud from which it was removed in step 3, then place a flatwasher, lockwasher, and 3/8" x 16 nut onto stud and tighten to 12-16 lb/ft. with a 9/16" wrench or socket.

b. Place brass washer (N401P43), voltage feedback wire, flatwasher, lockwasher, and 3/8" x 16 nut onto other mounting stud and tighten to 12-16 lb/ft. with a 9/16" wrench or socket.

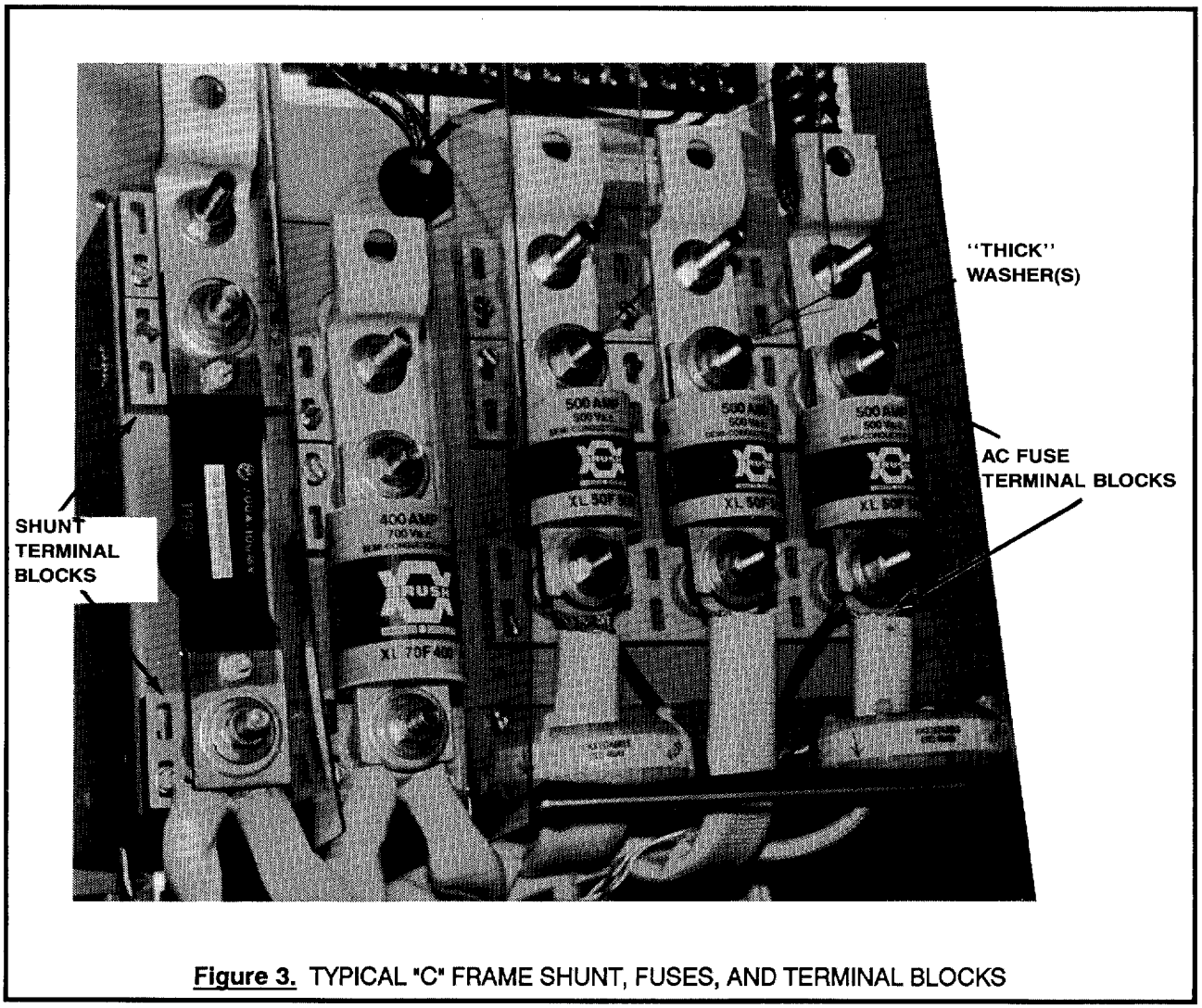


Figure 3. TYPICAL "C" FRAME SHUNT, FUSES, AND TERMINAL BLOCKS

"C" FRAME HORSEPOWER MODIFICATIONS

Shunt Changes (All) (See Figure 3)

WARNING - SHOCK AND BURN HAZARD:

Always disconnect power to the controller before changing any components of the controller. Failure to do so may cause serious injury to personnel and damage to the controller or the controlled equipment.

1. Verify that power to controller that shunt is to be changed is **OFF**.

2. Remove electrical connections from shunt as follows:
   a. Remove **control** connections by removing the two #10 nuts, lockwashers, and flatwashers from the control wire mounting studs of the shunt with a wrench or nutdriver.
   b. Remove **power** connections by removing the two 3/8" x 16 nuts, lockwashers, and flatwashers from the power cable mounting studs with a 9/16" wrench or socket.
   c. Lift shunt off from terminal block mounting studs.

3. Place new shunt (per Programming Software Package) over studs of terminal blocks (make sure flatwashers are still present under the shunt).

4. Replace electrical connections to shunt as follows:
   a. Orient all connectors to new shunt in same position as when removed in step 2 from old shunt.
   b. Place each control wire connector over shunt mounting stud and then place a #10 flatwasher, lockwasher, and nut onto stud and tighten to firmly with a wrench or nutdriver.
   c. Place power cable connector over shunt mounting stud and then place a flatwasher, lockwasher, and 3/8" x 16 nut onto stud and tighten to 12-16 lb/ft with a 9/16" wrench or socket.
   d. Place a flatwasher, lockwasher, and 3/8" x 16 nut onto other mounting stud and tighten to 12-16 lb/ft with a 9/16" wrench or socket.

AC and DC Fuse Changes (See Figure 3)

NOTE: *When changing* ***DC fuse*** *to 104X109AG015, the two fuse mounting terminal blocks must be changed to ones with 1/4" mounting studs (36C774302AAG17, includes hardware) and a brass washer (N401P41) must also be used when connecting voltage feedback wire.*

WARNING - SHOCK AND BURN HAZARD:

Always disconnect power to the controller before changing any components of the controller. Failure to do so may cause serious injury to personnel and damage to the controller or the controlled equipment.

1. Verify that power to controller that fuse(s) is to be changed is **OFF**.

2. Remove electrical connections from fuse by removing the two 3/8" x 16 nuts with a 9/16" inch wrench or socket.

3. Remove lockwashers, flatwashers, "thick" washers and power cable from studs and lift off fuse to be replaced.

4. Position new fuse (per Programming Software Package) over mounting studs to determine if terminal blocks must be moved (or changed for 104X109AG015 fuse) and proceed as follows: (if no movement or change is needed, proceed to step 5)

   If terminal blocks must be moved -
   a. Loosen and remove the #10 thread-forming mounting screws securing terminal blocks to frame and place new fuse onto the studs of terminal blocks (move blocks to accept new fuse).
   b. Position this assembly on frame to determine new mounting holes to be used (screws to be placed in outermost mounting holes only).
   c. Place a #10 thread-forming screw in outermost mounting holes of each terminal block and tighten to 55 lb/in. to secure terminal blocks to controller frame.
   d. Place flatwasher over each fuse terminal block stud.

If terminal blocks must be changed -

a. Loosen and remove the #10 thread-forming mounting screws securing terminal blocks to frame and remove these terminal blocks.

b. Retrieve new terminal blocks and mounting hardware and place a flatwasher over stud extending from each terminal block.

c. Place new fuse over studs of new terminal blocks and orient terminal blocks with mounting slots extending outward from ends of shunt.

d. Position this assembly on frame and determine mounting holes in frame to be used (screws to be placed in outermost mounting holes only).

e. Place a #8 thread-forming screw in outermost mounting holes of each terminal block and tighten to 28 lb/in. to secure terminal blocks to controller frame.

5. Make electrical connections to fuse as follows:

If terminal blocks were not changed -

a. Place power cable connector over fuse mounting stud from which it was removed in step 3, then place a flatwasher, lockwasher, and 3/8" x 16 nut onto stud and tighten to 12-16 lb/ft with a 9/16" wrench or socket.

b. Place a flatwasher, lockwasher, and 3/8" x 16 nut onto other mounting stud and tighten to 12-16 lb/ft with a 9/16" wrench or socket (voltage feedback wire and brass washer should still be present if DC fuse).

If terminal blocks were changed -

a. Place power cable connector over fuse mounting stud from which it was removed in step 3, then place a flatwasher, lockwasher, and 1/4" x 20 nut onto stud and tighten to 38-50 lb/in with a 7/16" wrench or socket.

b. Place brass washer (N401P41), voltage feedback wire, flatwasher, lockwasher, and 1/4" x 20 nut onto other mounting stud and tighten to 38-50 lb/in with a 7/16" wrench or socket.

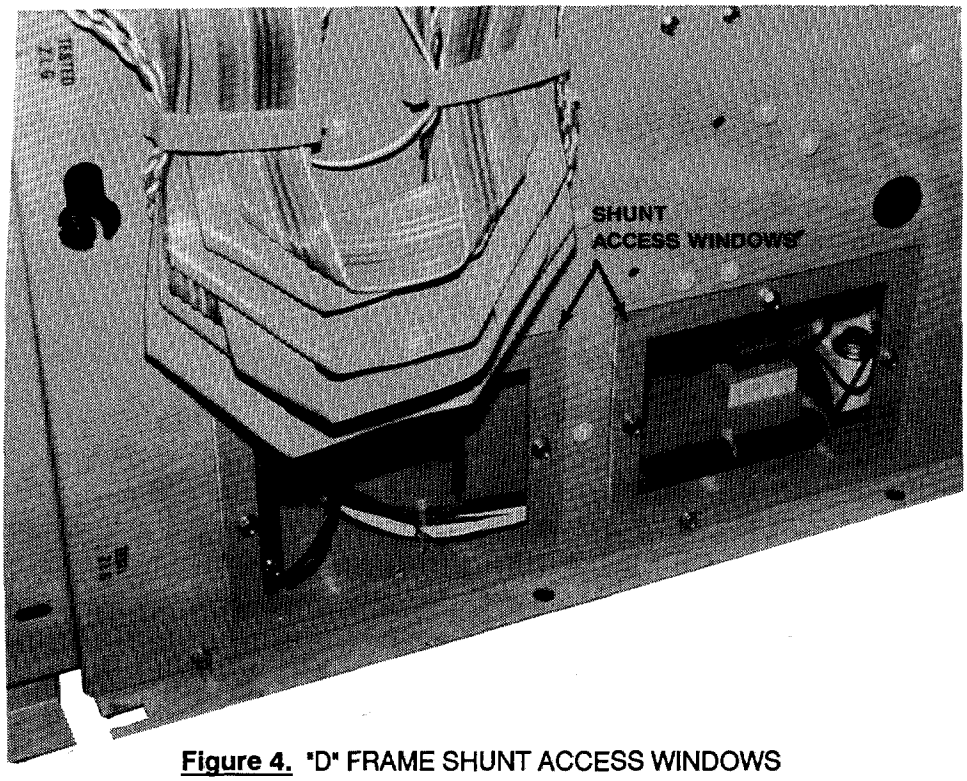


Figure 4. "D" FRAME SHUNT ACCESS WINDOWS

"D" FRAME HORSEPOWER MODIFICATIONS

Shunt Changes (All) (See Figure 4)

WARNING - SHOCK AND BURN HAZARD:

Always disconnect power to the controller before changing any components of the controller. Failure to do so may cause serious injury to personnel and damage to the controller or the controlled equipment.

1. Verify that power to controller that shunt is to be changed is **OFF**.

2. Remove the #10 thread-forming screws securing the shunt access window to the side of frame and remove access window.

3. Remove **control** connections by removing the two #10 nuts, lockwashers, and flatwashers from the control wire mounting studs of the shunt with a wrench.

4. Remove **power** connections and **shunt** as follows:
   a. Remove the two 1/2" x 13 nuts and flatwashers, from the bolts securing the shunt to the mounting brackets, using two 3/4" wrenches.
   b. Lift the shunt with bolts, flatwashers, lockwashers, and power cable out from the mounting brackets, then remove power cable from bolt.

5. Install new **shunt** (per Programming Software Package) and **power** connections to mounting brackets as follows:
   a. Insert the two 1/2 x 13" bolts with lockwashers, flatwashers, and power cable (in same position as when removed in step 4b) through the mounting holes of the new shunt and place this assembly onto the mounting brackets.
   b. Place a flatwasher and 1/2" x 13 nut onto each shunt mounting bolt extending through the mounting brackets and tighten to 29-39 lb/ft using two 3/4" wrenches.

6. Replace **control** connections to shunt as follows:
   a. Orient all connectors to new shunt in same position as when removed in step 3 from old shunt.
   b. Place each control wire connector over shunt mounting stud and then place a #10 flatwasher, lockwasher, and nut onto stud and tighten firmly with a wrench.

7. Reattach shunt access window to the frame by reinserting the #10 thread-forming screws removed in step 2 and tightening to 55 lb/in.

AC and DC Fuse Changes (See Figure 5)

NOTE: *300, 700, and 800 Horsepower controllers require **two DC fuses**. Mount second fuse in the same manner as the other fuse using the additional mounting hardware provided in the DC fuse bracket.*

WARNING - SHOCK AND BURN HAZARD:

Always disconnect power to the controller before changing any components of the controller. Failure to do so may cause serious injury to personnel and damage to the controller or the controlled equipment.

1. Verify that power to controller that fuse(s) is to be changed is **OFF**.

2. Remove electrical connections from fuse by removing the two 1/2" x 13 bolts from the clampnuts with a 3/4" inch wrench or socket.

3. Remove bolt, lockwashers, flatwashers, power cable, and clampnuts from brackets and remove fuse to be replaced.

4. Position new fuse (per Programming Software Package) onto mounting brackets to determine if terminal block/mounting bracket must be moved (place bolt through fuse and bracket to check) and proceed as follows: (if no movement is needed, proceed to step 5)

If AC fuse terminal blocks/mounting brackets must be moved -

a. Loosen and remove the two #10 thread-forming screws securing each top terminal block/bracket to frame and place new fuses onto the brackets (move **top terminal blocks/brackets only** to accept new fuses).

b. Place "locating" bolts through fuses and brackets to determine new mounting holes to be used and secure top brackets to frame with the #10 thread-forming screws removed in substep a and tighten to 55 lb/in..

If DC fuse mounting bracket must be moved -

a. Loosen and remove the 3/8" x 16 mounting bolts (from back of frame) securing fiber insulators to frame.

b. Place "locating" bolts through fuse and brackets to determine new mounting hole to be used.

c. Secure fiber insulators (in new location) to frame with the 3/8 x 16 bolts removed in substep b and tighten to 12-16 lb/ft..

d. Remove fuse and "locating" bolts from brackets.

5. Make electrical connections to fuse and install as follows:

   a. Place 1/2"x13 bolt (with lockwasher and flatwasher) through power cable connector, fuse, and bracket hole from which it was removed in step 3, then place clampnut underneath bracket hole, and tighten to 29-39 lb/ft with a 3/4" wrench or socket.

   b. Place 1/2"x13 bolt (with lockwasher and flatwasher) through fuse, and bracket hole from which it was removed in step 3, then place clampnut underneath bracket hole, and tighten bolt to 29-39 lb/ft with a 3/4" wrench or socket.

NOTE: *300, 700, and 800 Horsepower controllers require **two DC fuses**. Mount second DC fuse in the same manner as the other DC fuse using the additional mounting hardware provided in the DC fuse bracket.*

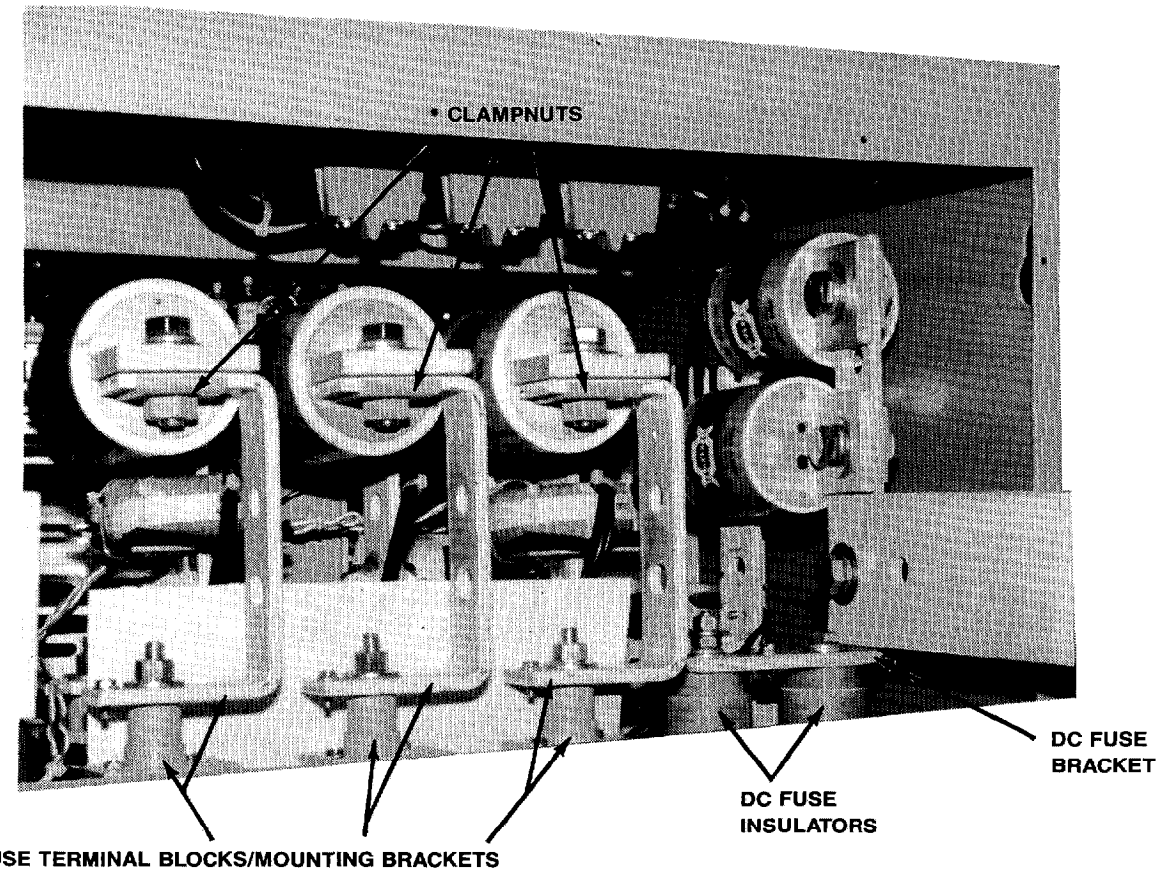


Figure 5. TYPICAL "D" FRAME FUSES, TERMINAL BLOCKS/MOUNTING BRACKETS, AND INSULATORS

CONTROL CARD HP SELECT POTENTIOMETER ADJUSTMENT

Whenever any shunts or fuses have been changed to modify the horsepower of the QIII DC Controller, the HP-SEL (P3) potentiometer on the Control card must be readjusted for the new components/horsepower. Use the following procedure and TABLE 2 to complete the necessary adjustments.

WARNING - SHOCK AND BURN HAZARD:
Always disconnect power to the controller before changing any components of the controller. Failure to do so may cause serious injury to personnel and damage to the controller or the controlled equipment.

1. Remove power from the QIII DC Controller.

2. Remove jumpers JP8, JP9, JP10, and JP11 from the Control card. (See Figure 6)

3. Measure the resistance (with a digital voltmeter [DVM]) between Control card test points "HS" and "ACOM".

4. Adjust potentiometer P3-HP SEL on the control card for the ohm value specified per module catalog number, motor horsepower, shunt catalog number, and DC voltage (see TABLE 2).

5. Replace jumpers JP8, JP0, JP10, and JP11 on the Control card (in positions called for per the Programming Software Package) after the appropriate resistance value has been set.

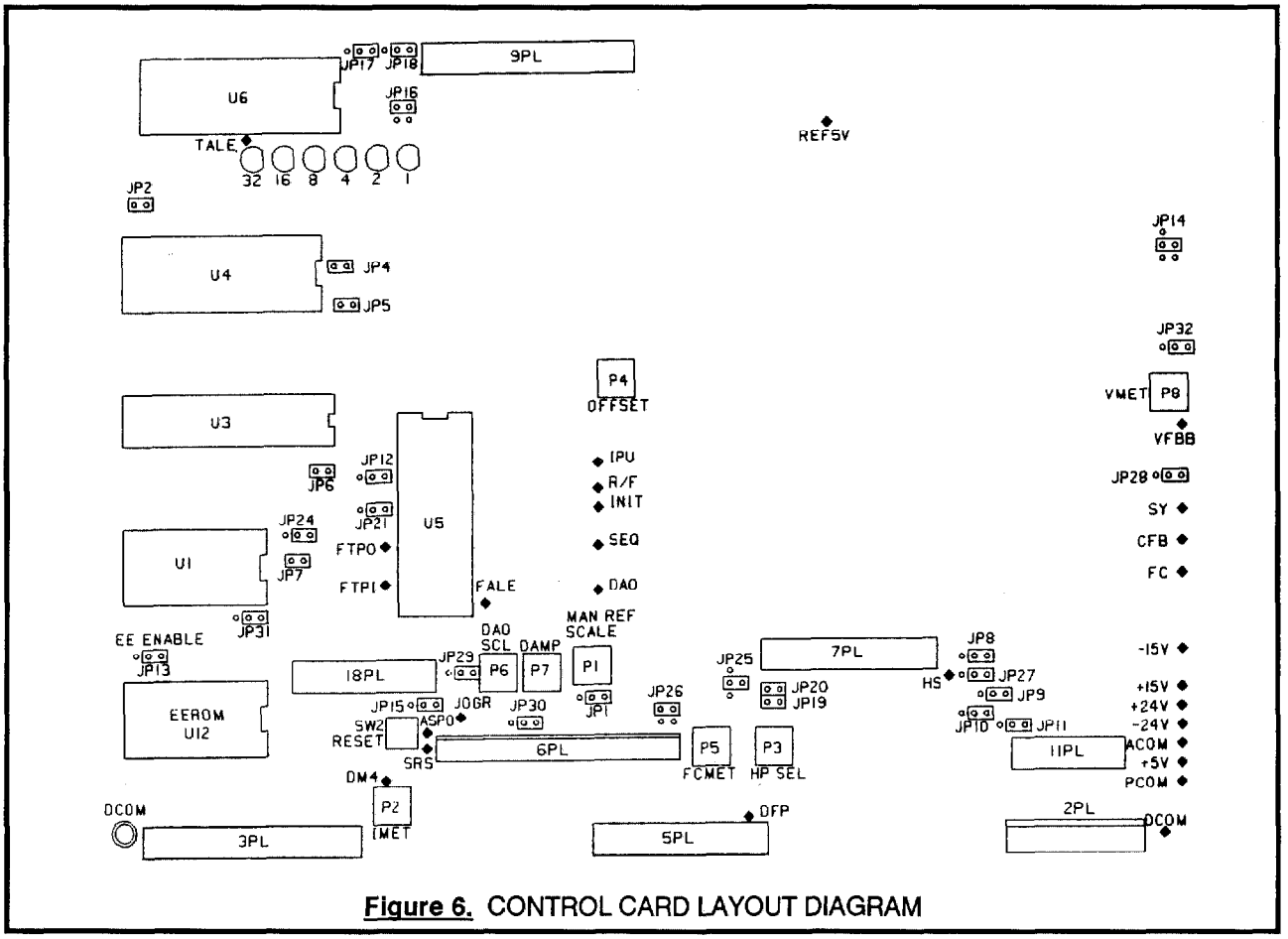


Figure 6. CONTROL CARD LAYOUT DIAGRAM

TABLE 2: CONTROL CARD P3-HP SEL RESISTANCE SETTINGS

Molule Catalog Number	HP AND Shunt Catalog Number at 240 VDC HP	104X149-	500 VDC HP	104X149-	P3-HP SEL Resistance Setting (HS to ACOM Test Points)
7VQD14	1	CC032	2	CC032	175.5
and	1.5	CC043	3	CC043	111.5
7VQD14R	2	CC033	5	CC033	52.5
	3	CC034	7.5	CC034	10.0
7VQD43	1	CC032	2	CC032	175.5
and	1.5	CC043	3	CC043	111.5
7VQD43R	2	CC033	5	CC033	52.5
	3	CC034	7.5	CC034	10.0
	5	CC035	10	CC035	210.0
	7.5	CC036	15	CC036	10.0
	10	CC042	20	CC042	198.5
	--	----	25	CC042	20.7
7VQD127	1	CC032	2	CC032	175.5
and	1.5	CC043	3	CC043	111.5
7VQD127R	2	CC033	5	CC033	52.5
	3	CC034	7.5	CC034	10.0
	5	CC035	10	CC035	210.0
	7.5	CC036	15	CC036	10.0
	10	CC042	20	CC042	198.5
	--	----	25	CC042	20.7
	15	CC037	30	CC037	93.4
	20	CC038	40	CC038	119.7
	25	CC039	50	CC038	60.9
	30	CC039	60	CC039	92.2
	--	----	75	CC040	10.0
7VQD260 and	40 & Below	CA008	--	----	90.0
7VQD260R	50	CA008	100 & Below	CA008	92.6
	60	CA009	125	CA009	68.3
	75	CA021	150	CA021	10.0

TABLE 2: CONTROL CARD P3-HP SEL RESISTANCE SETTINGS (continued)

Molule Catalog Number	HP AND Shunt Catalog Number at 240 VDC HP	104X149-	500 VDC HP	104X149-	P3-HP SEL Resistance Setting (HS to ACOM Test Points)
7VQD440 and 7VQD440R	40 & Below	CA008	--	----	10.0
	--	----	100 & Below	CA008	210.0
	50	CA008	--	----	177.5
	--	----	125	CA009	10.0
	60	CA009	--	----	127.5
	75	CA021	150	CA021	127.5
	--	----	200	CA010	210.0
	100	CA014	--	----	161.0
	--	----	250	CA014	210.0
	125	CA014	--	----	93.0
7VQD525 and 7VQD525R	40 & Below	CA008	--	----	10.0
	--	----	100 & Below	CA008	210.0
	50	CA008	--	----	177.5
	--	----	125	CA009	10.0
	60	CA009	--	----	127.5
	75	CA021	150	CA021	127.5
	--	----	200	CA010	210.0
	100	CA014	--	----	161.0
	--	----	250	CA014	210.0
	125	CA014	--	----	93.0
	--	----	300	CA011	210.0
	150	CA011	--	----	104.0
7VQD855 and 7VQD855R	200 & Below	CC007	400 & Below	CC007	210.0
	--	----	500	CC001	192.0
	250	CC001	--	----	90.0
7VQD1285 and 7VQD1285R	200 & Below	CC007	400 & Below	CC007	210.0
	--	----	500	CC001	192.0
	250	CC001	--	----	90.0
	--	----	600	CC002	14.0
	300	CC002	--	----	25.0
	--	----	700	CC003	50.0
	--	----	800	CC003	10.0

MFC/POWER SUPPLY CARD CFB (P3) POTENTIOMETER ADJUSTMENT

CFB ADJUST (P3) on the MFC/Power Supply card is set-up at the factory for the "standard" QIII DC Controller. Whenever the shunt has been changed to a different value other than what the controller was set-up for, the CFB ADJUST (P3) potentiometer on the MFC/Power Supply card must be readjusted for the new shunt value. Use the following procedure and Figure 7 to complete the necessary adjustment.

WARNING - SHOCK AND BURN HAZARD:

Always disconnect power to the controller before changing any components of the controller. Failure to do so may cause serious injury to personnel and damage to the controller or the controlled equipment.

1. Remove power from the QIII DC Controller.
2. Check and record the value of the new shunt (value is stamped on the face of the shunt).
3. Calculate the new CFB ADJUST setting using the following formula:
   CFB ADJUST = (Shunt Amps/IRATD) x 0.984
4. Apply power to the controller.
5. Set CFB ZERO potentiometer (P2) on the MFC/Power Supply card for less than (+/-)0.010 volts measured at the CFB Test Point on the Control card (set as close to zero as possible; use ACOM as reference common).
6. Remove power from the controller.
7. Move MFC/Power Supply card jumpers JP3 and JP4 to the 2-3 position and disconnect the DRPL plug connector from the MFC/Power Supply card.
8. Reapply power to the controller.
9. Set the CFB ADJUST potentiometer (P3) on the MFC/Power Supply card for the CFB ADJUST value calculated in step 3 (use a digital voltmeter measuring at CFB and ACOM Test Points on the Control card).
10. Return MFC/Power Supply card jumpers JP3 and JP4 to the 1-2 position and reconnect the DRPL plug connector to the MFC/Power Supply card.
11. Replace existing CFB calibration label:
    a. Obtain a blank CFB calibration label and mark it with the new CFB calibration volts.
    b. Remove old label from inside of cabinet.
    c. Attach new label (with new CFB calibration volts marked) to inside right-hand of cabinet just above the MFC/Power Supply card.

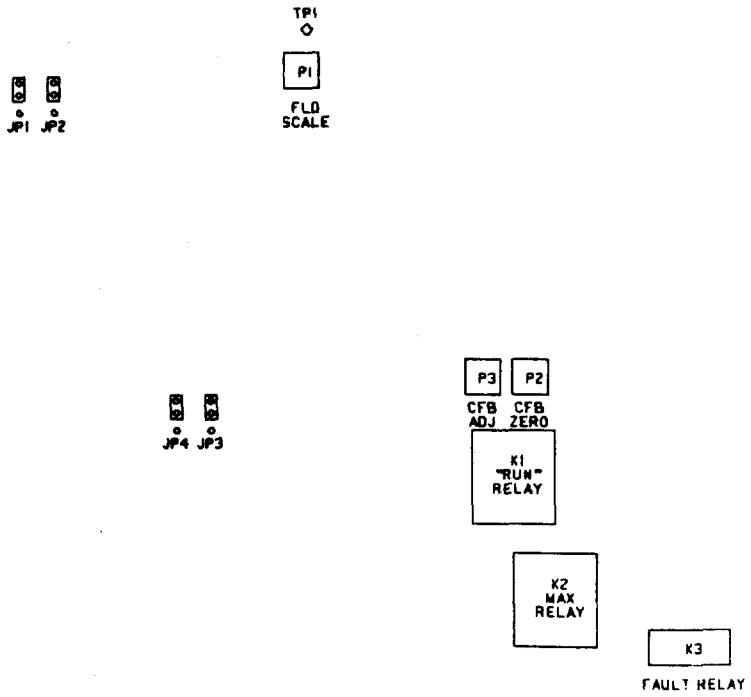


Figure 7. MFC/POWER SUPPLY CARD LAYOUT DIAGRAM

NOTES:



GE Drive Systems

Preliminary GEK-45166 (10/90)

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